

GeezNet: Comparative Study of Deep Architectures for Geez Numeral Recognition

^{1st} Oli Gurmesssa

*Department of Computer and Data Sciences
University of Saint Thomas
Minnesota, USA
gurm1658@stthomas.edu*

^{2nd} Pakeeza Akram

*Department of Computer and Data Sciences
University of Saint Thomas
Minnesota, USA
akra6203@stthomas.edu*

Abstract—The Ge’ez script is one of the ancient writing systems. It is underrepresented in modern OCR research due to its complex numeral shapes and the absence of standardized datasets. This paper presents GeezNet, a deep learning framework optimized for Ge’ez numeral recognition. We also introduce a curated, open-access dataset containing 10,000 labeled images across 20 numeral classes. Several GeezNet variants—baseline, depthwise separable, residual, and Vision Transformer (ViT) hybrid—are compared with transfer learning models such as ResNet-50 and Inception-V3. The optimized 15-layer GeezNet achieves 97.8% accuracy on an independent test set. It reduces computational cost compared to larger generic architectures, enabling deployment on mobile and archival digitization systems. Misclassification analysis shows difficulties with visually similar numerals. These insights inform future work on robustness to noise and handwriting variability. This study provides a benchmark model, a comprehensive architecture comparison, and a public dataset to advance OCR for underrepresented scripts in low-resource settings.

Index Terms—ancient script digitization, convolutional neural networks (CNN), deep learning, Ge’ez numeral recognition, GeezVision, heritage AI, low-resource languages, optical character recognition (OCR)

I. INTRODUCTION

The Ge’ez script, also known as Classical Ethiopic, is one of the oldest written languages still in use today. Originating in the fifth century BCE, Ge’ez has profoundly shaped Ethiopia’s linguistic heritage. Today, it remains central to the liturgical traditions of the Ethiopian and Eritrean Orthodox churches [1] and has left a lasting imprint on modern languages such as Amharic and Tigrinya.

Despite its rich cultural legacy, Ge’ez remains underrepresented in digital tools and modern Optical Character Recognition (OCR) systems. Thousands of historical documents inscribed in this script are vulnerable to degradation due to age and environmental conditions. Digitizing these documents is not just a technical pursuit; it is crucial for preserving the linguistic and cultural heritage of the region.

The Ge’ez numeral system adds further complexity to this challenge. It comprises 20 unique numeric symbols. It includes characters for units (1–9), tens (10–90), 100, and 10,000, but notably lacks a representation for zero. These numerals are highly stylized and visually distinct from commonly used numerals, posing significant challenges for automated recognition

systems. Furthermore, the scarcity of standardized datasets and the ornate nature of the script increase the technical hurdles in developing effective OCR solutions.

A. Challenges in Recognizing Ge’ez Numerals Digitally

Recognizing Ge’ez numerals presents unique challenges that set them apart from more widely studied numeral systems.

- The symbol set is visually complex, with 20 distinct, ornate characters. Many of these share similar geometric patterns, increasing the likelihood of misclassifications, especially in handwritten forms.
- There is a lack of publicly available datasets. Researchers often rely on proprietary or synthetic datasets that vary in quality and formatting.
- Variability in handwriting and typography adds difficulty. Differences in stroke pressure, slant, and writing style reduce model generalization, especially in low-resource scripts where diverse data is difficult to obtain.
- Many Ge’ez manuscripts are physically degraded, exhibiting smudges, faded ink, or torn pages. Such artifacts introduce noise and complicate OCR pipelines, even for advanced deep learning models.

Together, these challenges make the development of reliable OCR systems for Ge’ez numerals particularly demanding.

B. Literature Review

Early efforts in Ge’ez numeral recognition primarily relied on traditional image processing techniques tailored for historical manuscripts. [2] developed a custom segmentation approach for ancient Ge’ez manuscripts written on vellum, combining adaptive noise filtering, global thresholding, and stroke-level segmentation to achieve 79.32% recognition accuracy using a multiclass SVM. While effective for degraded historical texts, such methods were limited in scalability and generalization.

In related domains, similar OCR challenges have been addressed for other underrepresented scripts such as Amharic [3], Tigrinya [4], and Sinhala [5]. These studies highlight the importance of script-specific feature extraction to overcome morphological complexity. For instance, [3] applied CNN-based recognition to Amharic handwritten characters, achieving promising results despite limited datasets. [4] proposed a

Tigrinya OCR pipeline using handcrafted features combined with SVM classification. The paper highlighted the need for deeper architectures to handle script variability. [5] represents the potential of deep learning to achieve state-of-the-art accuracy for Sinhala script. It also emphasized the trade-off between model size and deployability in resource-constrained environments.

With the rise of deep learning, researchers began leveraging CNN architectures for printed numerals. [6] trained a CNN on a synthetically generated dataset of Ge'ez numerals, achieving 97.29% accuracy on clean, uniform data. Comparable efficiency-oriented CNN architectures like MobileNet [7] and EfficientNet [8] have demonstrated that parameter reduction is possible without compromising accuracy, which aligns with the design philosophy behind GeezNet.

To address the need for authentic data, [9] introduced a large-scale handwritten dataset containing over 51,000 digits, showing a performance gap between synthetic and real-world handwriting. Similar approaches have been taken with benchmark datasets like MNIST [10] and EMNIST [11], which standardize preprocessing and enable fair model comparisons.

Building on this, [12] explored sequential modeling using a CNN-LSTM hybrid, improving recognition of flowing scripts but increasing computational complexity. Comparable sequence-based recognition methods have been applied to offline handwriting recognition for scripts like Arabic [13] and Devanagari [14], underscoring the universality of temporal modeling challenges.

Recent advancements in Vision Transformers (ViTs) have expanded OCR capabilities beyond CNNs. [15] displayed that pure transformer architectures can achieve competitive results in image classification. In contrast, [16] applied transformer-based models to text recognition in scene images, improving robustness to distortions and irregular layouts. Integrating CNN feature extraction with transformer encoders has emerged as a promising hybrid approach for OCR in heritage preservation contexts.

Collectively, the literature indicates three key trends: (1) domain-specific model design yields superior results compared to generic architectures; (2) efficient networks enable deployment in low-resource settings without significant accuracy loss; and (3) hybrid CNN-transformer models are becoming increasingly relevant for complex, ornate scripts like Ge'ez. However, there remains a scarcity of openly available, high-quality datasets for underrepresented scripts, limiting reproducibility and large-scale benchmarking. GeezNet directly addresses these gaps by combining an optimized architecture with a publicly released handwritten Ge'ez numeral dataset.

While these studies collectively demonstrate the feasibility of OCR for low-resource scripts, there remains a disconnect between state-of-the-art models and practical deployment in resource-constrained environments. Most existing solutions either rely on synthetic datasets or use architectures that are too computationally heavy for field digitization efforts. This gap underscores the need for an efficient, purpose-built framework like GeezNet, coupled with a publicly accessible, diverse

dataset.

C. OCR and the Preservation of Cultural Heritage

Optical Character Recognition (OCR) technologies have emerged as essential tools for preserving linguistic and cultural heritage, especially for low-resource and historically significant scripts. Digitization of ancient texts enables searchable archives, academic access, and revitalization efforts in endangered and liturgical languages [17]. The Ge'ez script, in particular, holds immense historical, religious, and cultural value in Ethiopian and Eritrean traditions, yet remains under-represented in OCR research and digital libraries [18].

Efforts such as the OCR4all project [19] and initiatives targeting South Asian scripts [20] have demonstrated the importance of tailored OCR systems in handling complex, ornate, or degraded scripts. These works emphasize that generic OCR systems often fail to capture the nuances of script-specific morphology, particularly in low-resource settings.

By introducing GeezNet, a purpose-built deep learning architecture for Ge'ez numeral recognition, we aim to bridge this gap. The curated dataset and the model proposed in this study serve as foundational components for developing comprehensive OCR pipelines. These pipelines could be integrated into manuscript digitization workflows, supporting cultural institutions, scholars, and language communities engaged in preserving Ge'ez heritage. In doing so, this research not only advances the technical frontier of OCR but also contributes to the digital preservation of an ancient and culturally significant script.

D. Motivation and Contributions

Digitizing the Ge'ez script is essential for preserving the linguistic and cultural heritage of Ethiopia and Eritrea. Yet, reliable OCR solutions remain scarce due to limited datasets and the ornate numeral system. This study addresses the following key gaps:

- Lack of practical, efficient models: Most existing approaches depend on synthetic data or require large, resource-intensive architectures, limiting their applicability for real-world deployment.
- Data scarcity: There is a shortage of well-curated, diverse, publicly available handwritten datasets for Ge'ez numerals, hindering progress and reproducibility.
- Accuracy-efficiency trade-off: Previous work reveals a trade-off between model performance and computational efficiency, highlighting the need for architectures that optimize both for mobile and heritage applications.

To address these gaps, we make the following contributions:

- We introduce and publicly release a balanced dataset of 10,000 handwritten Ge'ez numeral images, spanning 20 classes and capturing natural handwriting variability.
- We design and systematically evaluate a family of custom deep learning models for Ge'ez numeral recognition, including an 8-layer CNN, the deeper 15-layer GeezNet, a depthwise separable variant (GeezNet-DSep), a residual variant (GeezNet-Res), and a Vision Transformer hybrid

(GeezNet-ViT). All are optimized for 28×28 grayscale images.

- We conduct a thorough comparative study of all GeezNet variants against transfer learning models (ResNet-50 and InceptionV3), evaluating classification accuracy, parameter count, computational efficiency, and generalization.
- Our optimized 15-layer GeezNet achieves 97.8% test accuracy, outperforming all other models—including InceptionV3 and ResNet50—while using dramatically fewer parameters and memory, making it ideal for mobile and archival OCR.

These contributions advance the state of OCR for underrepresented scripts by providing robust models, a high-quality open dataset, and a thorough comparative analysis, laying the foundation for future research in low-resource document analysis and heritage preservation. Beyond the Ge'ez numeral system, the methodological insights from this study—particularly in balancing accuracy and efficiency—are directly transferable to OCR tasks for other underrepresented scripts such as N'Ko, Tifinagh, and Brahmi, enabling broader adoption in global heritage digitization projects.

II. METHODOLOGY

The development and evaluation pipeline for GeezNet included dataset construction, preprocessing, augmentation, feature analysis, model design, and evaluation strategy, as illustrated in Fig. 1.

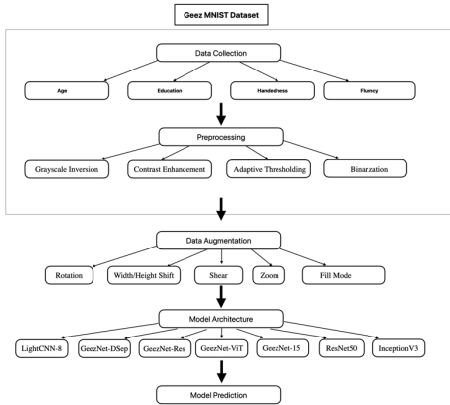


Fig. 1. Workflow of the Ge'ez numeral recognition pipeline, illustrating the stages from data collection and preprocessing to augmentation, model training (LightCNN-8, GeezNet variants, ResNet50, InceptionV3), and prediction.

A. Dataset

A handwritten Ge'ez numeral dataset was curated, comprising 10,000 labeled samples across 20 classes. Each sample was written on blank paper using dark-tipped markers. Data were collected from over 150 individuals (ages 18–24), representing diverse handwriting styles. After scanning, sheets were segmented with bounding boxes and labels verified manually to minimize noise. The dataset was evaluated using stratified 5-fold cross-validation for robust performance assessment. The

complete dataset is publicly available at [21] to promote reproducibility.

B. Data Preprocessing

All images underwent preprocessing steps to ensure uniformity and enhance signal quality. RGB images were converted to grayscale, then binarized with adaptive thresholding and denoised with Gaussian filtering ($\sigma = 3$). Cropping centered the numeral, and images were resized to 28×28 pixels, standardizing input dimensions. This pipeline followed established methods for handwritten digit datasets (e.g., MNIST). Examples are shown in Fig. 2.

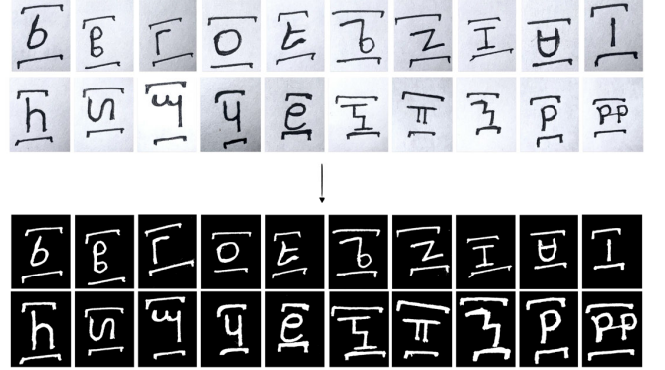


Fig. 2. Sample Ge'ez numeral images before (top) and after (bottom) preprocessing, including cropping, binarization, and grayscale conversion.

C. Image Augmentation

To enhance generalization, controlled geometric transformations were applied: random rotations ($\pm 10^\circ$), width and height shifts (up to 10%), zoom ($\pm 10\%$), and shear. Nearest-neighbor interpolation was used to preserve digit structure. Horizontal/vertical flipping was excluded due to the asymmetry of Ge'ez numerals, which could introduce label ambiguity.

D. Feature Visualization (t-SNE)

The quality and structure of the dataset were further evaluated with t-distributed stochastic neighbor embedding (t-SNE), projecting learned CNN features into two dimensions. Fig. 3 shows distinct clusters for all 20 numeral classes, confirming effective feature learning and inter-class separability. Minor overlap was observed between visually similar classes, such as Class 12 and Class 13 (e.g., digits 40 and 50), while others such as Class 19 (e.g., digit 10,000) and Class 17 (digit 90) formed isolated clusters.

E. Deep Learning Architectures

Six architectures were implemented: two baseline CNNs (8-layer, 15-layer), a depthwise separable CNN, a residual CNN, a Vision Transformer hybrid (ViT Hybrid), and two transfer learning models (ResNet50 and InceptionV3). All models were trained with stratified 5-fold cross-validation, 20 epochs, the Adam optimizer (learning rate = 0.001), and categorical cross-entropy loss.

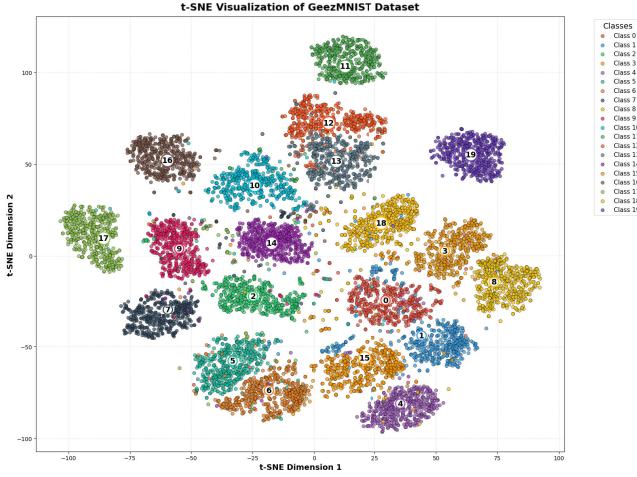


Fig. 3. t-SNE visualization of CNN feature space for Ge’ez numerals. Clusters correspond to individual numeral classes.

1) *LightCNN-8*: The 8-layer CNN is a compact baseline, optimized for speed and resource efficiency. It uses two convolutional layers (32 and 64 filters) with ReLU activation and max pooling, a dense layer (128 units), dropout (0.5), and a softmax output for 20 classes.

2) *GeezNet-15*: This 15-layer CNN consists of three convolutional blocks (32, 64, 128 filters) with batch normalization, ReLU, and max pooling. Features are flattened and passed to a 256-unit dense layer with dropout, followed by a softmax output. The model captures complex patterns with low risk of overfitting.

3) *GeezNet-DSep*: This variant uses depthwise separable convolutions (three blocks: 32, 64, 128 filters), batch normalization, and max pooling. The output is flattened, followed by a dense layer (256 units), dropout, and a softmax layer. It reduces parameters and computational load.

4) *GeezNet-Res*: The residual architecture introduces short-cut connections to improve gradient flow and learning. Convolutional layers (32, 64, 128 filters), batch normalization, and custom residual blocks are used, followed by dense and softmax layers.

5) *GeezNet-ViT*: The ViT Hybrid merges convolutional layers (32 and 64 filters) and max pooling with patch embedding and a compact transformer block. The sequence is flattened and passed to a dense layer (256 units), dropout, and softmax output. This approach draws inspiration from the broader Vision Transformer literature [15], which has shown strong results in image classification tasks and recently in OCR pipelines [16].

6) *ResNet50 (Transfer Learning)*: ResNet50 is a deep residual network, trained from scratch on Ge’ez numerals. Flattening, a 256-unit dense layer, dropout, and a softmax output follow the base network. High computational requirements make it less suited for low-resource settings.

7) *InceptionV3 (Transfer Learning)*: InceptionV3 employs multi-path convolutions for multi-scale feature extraction. The network output is flattened and passed to a dense layer (256

units), dropout, and softmax. It requires substantial resources, best suited for high-capacity environments.

F. Ablation Study

To assess the contribution of each architectural component, we conducted an ablation study starting from the LightCNN-8 baseline and incrementally adding complexity:

- **Depth Increase**: Expanding to 15 layers (GeezNet-15) improved accuracy by 2.9% while increasing $4.5\times$ parameters.
- **Depthwise Separable Convolutions**: Replacing standard convolutions reduced parameters by 83.5% with only a 0.8% drop in accuracy.
- **Residual Connections**: Adding skip connections improved gradient flow, boosting accuracy by 1.1% over the baseline while keeping parameters low.
- **ViT Hybridization**: Incorporating a transformer encoder improved performance on visually similar classes, as confirmed by confusion matrix analysis.

Table I summarizes the results.

TABLE I
ABLATION STUDY: ACCURACY VS. PARAMETERS

Model	Params	Accuracy (%)	Δ from Baseline
LightCNN-8	423k	94.9	—
GeezNet-15	1.9M	97.8	+2.9
GeezNet-DSep	313k	94.1	-0.8
GeezNet-Res	487k	96.0	+1.1
GeezNet-ViT	325k	96.8	+1.9

A summary of model architectures and evaluation strategy is provided in Table II.

TABLE II
SUMMARY OF MODEL ARCHITECTURES AND EVALUATION STRATEGY

Model	Type	Depth	Input	Params	Key Features
LightCNN-8	Baseline CNN	8	$28\times 28\times 1$	423k	2 Conv, 1 Dense, Dropout
GeezNet-15	Custom CNN	15	$28\times 28\times 1$	1.9M	3 Conv blocks, BatchNorm
GeezNet-DSep	Depthwise CNN	10	$28\times 28\times 1$	313k	SeparableConv, BatchNorm
GeezNet-Res	Residual CNN	12	$28\times 28\times 1$	487k	Residual, BatchNorm
GeezNet-ViT	Conv+ViT Hybrid	12	$28\times 28\times 1$	325k	PatchEmbed, Transformer
ResNet50	Deep ResNet	50	$32\times 32\times 3$	23.6M	Residual, Deep, Pretrained
InceptionV3	Inception	48	$75\times 75\times 3$	21.8M	Multi-path

III. RESULTS AND DISCUSSIONS

A comprehensive evaluation was conducted using six deep learning models for Ge’ez numeral recognition: two baseline CNNs (8-layer, 15-layer GeezNet), depthwise separable (GeezNet-DSep), residual (GeezNet-Res), Vision Transformer hybrid (GeezNet-ViT), and transfer learning baselines (InceptionV3 and ResNet50). All models were implemented in Python (Keras/TensorFlow) and trained on Google Colab GPUs with stratified 5-fold cross-validation. Each model was trained for 20 epochs using the Adam optimizer and categorical cross-entropy loss. ResNet50, which consistently failed to surpass 50% accuracy, is excluded from detailed comparative analysis.

A. Performance Comparison

Table III and Fig. 4 demonstrate that GeezNet-15 achieved the highest mean F1 score (0.977), precision (97.75%), recall (97.70%), and ROC-AUC (99.97%). Other custom models—including ViT Hybrid, Residual, and DSep—also delivered strong performance (F1 > 0.91), while lightweight models like LightCNN-8 offered efficiency with slightly lower accuracy. Transfer learning baselines underperformed on this domain-specific task.

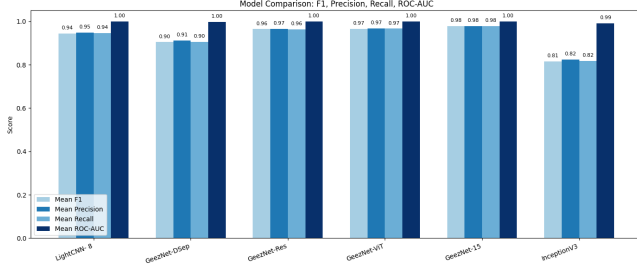


Fig. 4. Comparison of mean F1, precision, recall, and ROC-AUC for all models.

TABLE III
5-FOLD CROSS-VALIDATION PERFORMANCE METRICS

Model	F1	Precision	Recall	ROC-AUC
LightCNN-8	0.944	94.7%	94.5%	99.9%
GeezNet-DSep	0.904	91.1%	90.4%	99.8%
GeezNet-Res	0.964	96.5%	96.4%	99.9%
GeezNet-ViT	0.966	96.7%	96.6%	100.0%
GeezNet-15	0.977	97.8%	97.7%	100.0%
InceptionV3	0.815	82.4%	81.7%	99.0%

B. Impact of Architectural Modifications

The performance differences observed between LightCNN-8, GeezNet-15, and its variants align with the ablation findings in Section I. Increasing depth from 8 to 15 layers delivered the largest performance gain (+2.9% accuracy) by allowing the network to capture more intricate stroke-level features characteristic of ornate Ge'ez numerals. Depthwise separable convolutions, while drastically reducing parameter count (−83.5%), produced only a minor drop in accuracy, highlighting their suitability for extreme resource-constrained deployments. Residual connections improved gradient stability and slightly enhanced recall, particularly for classes prone to confusion in Fig. 11. Finally, ViT hybridization improved disambiguation between visually similar numerals such as 40 and 50, consistent with transformer-based global context modeling.

Overall, GeezNet-15 strikes the most effective balance between representational depth, stability, and computational efficiency. Its additional convolutional layers allow for progressive abstraction of low-level strokes into high-level structural patterns unique to the Ge'ez numeral system, yielding both high accuracy and robust generalization across diverse handwriting styles, while remaining deployable in practical OCR scenarios.

C. Generalization and Test Set Performance

Table IV presents both the mean accuracy from 5-fold cross-validation and the test set accuracy, where the latter was computed on a separate hold-out dataset of 2,039 images not used during training or validation. GeezNet-15 achieved the highest test accuracy (97.8%), confirming its strong generalization to unseen data. Other custom models—particularly the residual and ViT variants—also showed robust generalization performance. Lightweight models offered greater efficiency but with some reduction in accuracy, while InceptionV3, despite its large capacity, underperformed relative to the best custom architectures.

TABLE IV
MEAN CROSS-VALIDATION ACCURACY (5-FOLD) AND HOLD-OUT TEST SET ACCURACY ($N=2,039$) FOR EACH MODEL.

Model	Mean Accuracy	Std	Test Accuracy
LightCNN-8	94.5%	0.0057	94.3%
GeezNet-DSep	90.4%	0.0229	94.1%
GeezNet-Res	96.4%	0.0037	97.4%
GeezNet-ViT	96.6%	0.0056	95.5%
GeezNet-15	97.7%	0.0027	97.8%
InceptionV3	81.6%	0.0161	91.4%

D. Computational Efficiency

Resource utilization is summarized in Table V. GeezNet-15 achieved state-of-the-art accuracy with moderate storage and runtime requirements, significantly outperforming larger transfer learning models in memory and speed. This makes GeezNet-15 suitable for deployment in resource-limited environments.

TABLE V
COMPUTATIONAL COST AND MEMORY REQUIREMENTS

Model	Train Time (s)	Infer Time (s)	Memory (MB)
LightCNN-8	0.0003	0.004	6.0
GeezNet-DSep	0.0003	0.006	118
GeezNet-Res	0.0004	0.008	157
GeezNet-ViT	0.0004	0.008	195
GeezNet-15	0.0004	0.011	174
InceptionV3	0.0023	0.065	1,883

E. Learning Dynamics and Model Stability

Learning curves (Figs. 5–10) indicate that GeezNet-15, GeezNet-Res, and GeezNet-ViT converge rapidly with stable validation accuracy, reflecting strong generalization and minimal overfitting. Lightweight models (LightCNN-8, GeezNet-DSep) are efficient but display slightly larger gaps between training and validation accuracy. InceptionV3 shows slower, less stable convergence, emphasizing the advantage of domain-specific architectures.

F. Confusion Matrix and Error Analysis

After cross-validation, each model was retrained on the whole training set and evaluated on the hold-out test set.

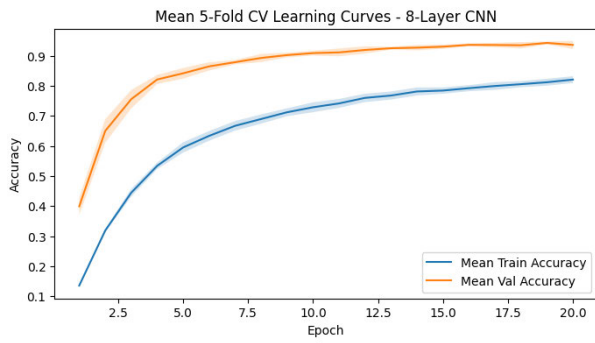


Fig. 5. 8-layer CNN learning curve showing steady accuracy increase and consistent loss decline.

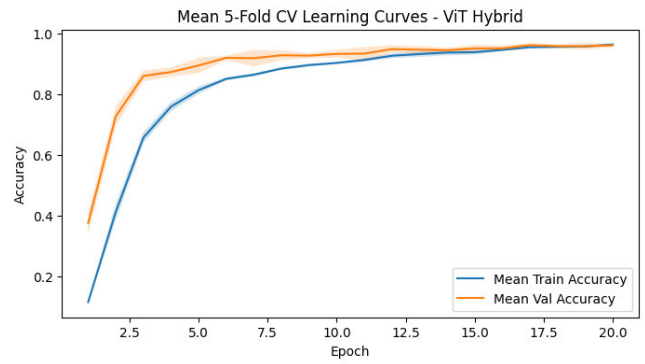


Fig. 9. GeezNet-ViT Hybrid exhibits gradual, stable convergence with accuracy over 95%.

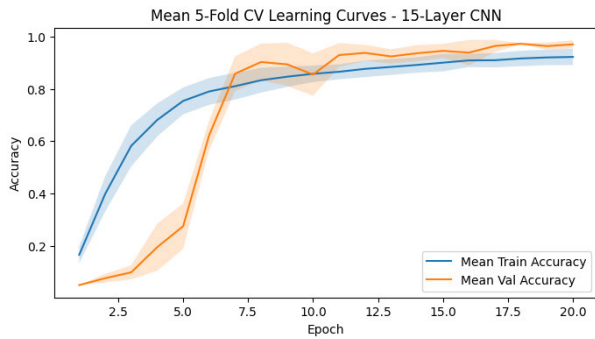


Fig. 6. GeezNet (15-layer CNN) shows high accuracy and stable validation trends.

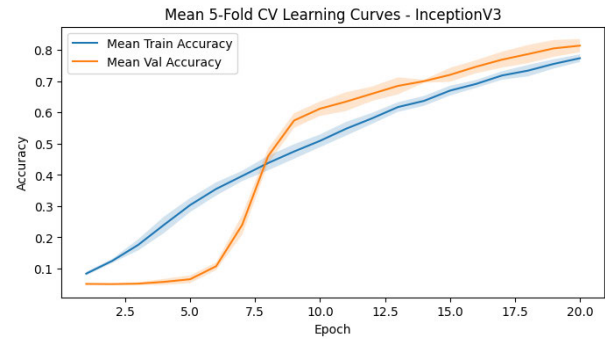


Fig. 10. InceptionV3 learning curves showing gradual and stable convergence, with validation accuracy plateauing below custom models.

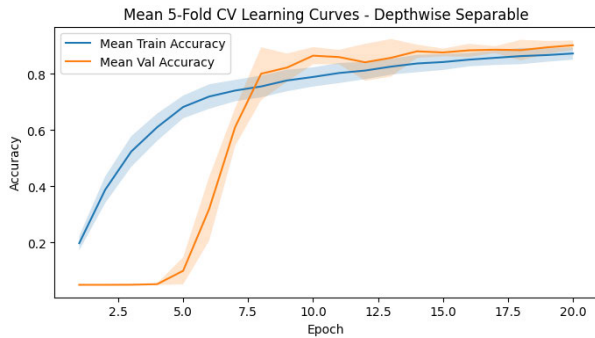


Fig. 7. GeezNet-DSep shows steady but slower convergence

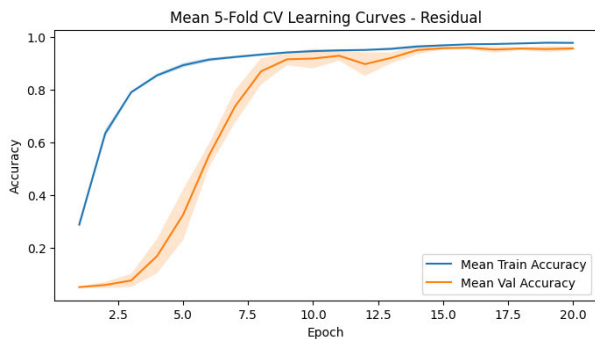


Fig. 8. GeezNet-Res shows high accuracy and stable validation trends.

Fig. 11 shows the confusion matrix for GeezNet-15, demonstrating strong diagonal dominance and high class-wise accuracy. Occasional off-diagonal entries reflect rare misclassifications among visually similar numerals.

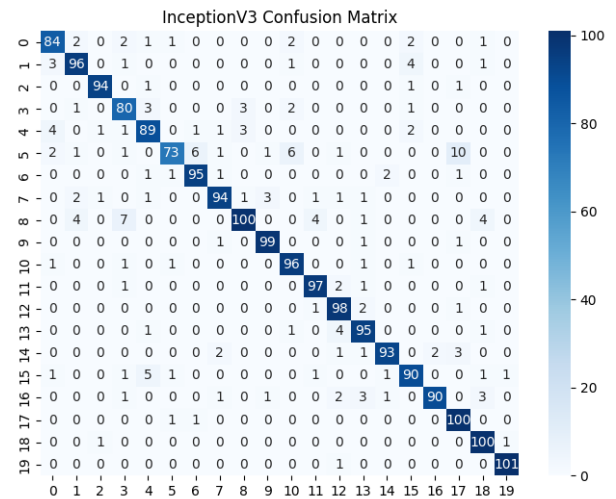


Fig. 11. Confusion matrix for GeezNet-15 on the test set.

G. Error Patterns

Systematic misclassifications are observed between digits with visually similar handwritten forms, as shown in the confusion matrix. For instance, class 3 (digit 4) is often confused with classes 5, 6, and 8, while class 13 (digit 40) occasionally overlaps with class 10 (digit 100). Such errors are primarily attributed to ambiguous or degraded glyph shapes, which lead to confusion between numeral pairs with overlapping visual features. Despite these challenges, the matrix remains strongly diagonal, indicating robust overall classification performance.

A closer inspection of the confusion matrix in Fig. 11 reveals that most misclassifications occur between numerals sharing similar curvilinear components or stroke angles. For example, digits 40 and 50 differ primarily in a minor curvature extension, which becomes ambiguous in handwritten samples with degraded ink or faint strokes. Similarly, digits 4, 6, and 8 share a typical loop structure, making them susceptible to confusion when written in a condensed style. This suggests that future work could benefit from integrating stroke-width normalization or attention-based modules to capture subtle shape deviations.

H. Discussion

The results indicate that compact, task-specific CNNs, such as GeezNet, deliver both high accuracy and strong computational efficiency. In contrast, deeper generic architectures [7], [8] require substantially more parameters and memory, yet offer no performance advantage. GeezNet therefore provides a lighter and more practical solution without sacrificing effectiveness.

Similar efficiency-accuracy trade-offs have been reported in heritage digitization projects for scripts like Sinhala [5] and Arabic [13], reinforcing the applicability of GeezNet’s design to low-resource domains.

I. Model Size and Deployment Considerations

As shown in Table VI, we compare model size and parameter count. GeezNet-15 achieves the best balance of accuracy and efficiency. The transfer learning models require significantly more memory with no performance benefit.

While InceptionV3 contains over 21M parameters and occupies 83.3 MB, GeezNet-15 achieves superior accuracy with only 1.9M parameters and a 7.3 MB footprint—a 90% reduction in model size. This lightweight profile enables deployment on devices with limited memory, such as mobile phones or Raspberry Pi boards, without compromising recognition quality. Fig. 4 further illustrates that parameter count does not directly translate to accuracy, underscoring the benefits of task-specific model design.

J. Summary

GeezNet-15 provides state-of-the-art accuracy and efficiency for Ge’ez numeral recognition. Custom models consistently outperform transfer learning baselines, confirming the effectiveness of domain-specific, mid-sized CNNs for OCR in

TABLE VI
MODEL SIZE AND PARAMETER COMPARISON

Model	Depth	Size (MB)	Parameters
CNN-8	8	1.6	423k
GeezNet-15	15	7.3	1.9M
GeezNet-DSEp	10	1.2	313k
GeezNet-Res	13	1.9	487k
GeezNet-ViT	12	1.2	325k
InceptionV3	48	83.3	21.8M

low-resource settings. These results validate the practical value of GeezNet and its variants for cultural heritage digitization.

Overall, these results confirm that designing architectures tailored to the unique morphological complexity of underrepresented scripts is more effective than reusing generic large-scale models. GeezNet-15 balances accuracy, memory efficiency, and generalization, making it suitable for both archival digitization and on-device recognition. The comparative study also highlights that efficient architectural choices, such as depthwise separable convolutions and residual connections, can maintain high accuracy while drastically lowering computational demands, opening the path for OCR solutions in other low-resource languages.

IV. CONCLUSION

This work introduced GeezNet—a family of deep learning models optimized for handwritten Ge’ez numeral recognition. It addresses a critical gap in OCR research for underrepresented and culturally significant scripts. Using a curated dataset of 10,000 labeled images across 20 numeral classes, we evaluated multiple GeezNet variants against both lightweight baselines and large-scale transfer learning models.

The optimized 15-layer GeezNet achieved 97.7% cross-validation and 97.8% test accuracy. It maintains a compact size of just 1.9 million parameters and a 7.3 MB memory footprint—over 90% smaller than InceptionV3, yet more accurate. These results confirm that task-specific architectures can outperform generic models when tuned for domain characteristics.

Ablation studies revealed that architectural depth significantly improves performance. Moving from 8 to 15 layers led to a 2.9% increase in accuracy. Depthwise separable convolutions reduced model size by 83.5% with minimal performance loss. Residual connections helped stabilize training and improve recall for ambiguous digits. Vision Transformer hybrids further enhanced accuracy by modeling global context, especially for curvilinear numeral pairs.

Despite its compactness, GeezNet outperformed larger models like InceptionV3. This finding reinforces that parameter count alone does not guarantee better performance. Instead, architecture tuned to the target script’s morphology yields greater efficiency and scalability. GeezNet is well-suited for real-world deployment on a variety of platforms, including mobile devices, cloud systems, and embedded units like the Raspberry Pi.

While the results are promising, several directions remain open. The current dataset consists of clean, uniformly processed samples. However, real-world manuscripts often contain noise, ink bleed-through, faded characters, and damage due to aging. Future work will focus on improving robustness under such conditions. We aim to expand the dataset by including samples from degraded manuscripts and low-light environments. In parallel, we will introduce synthetic noise and distortion during training to simulate real-world challenges. Additionally, we plan to integrate GeezNet into full OCR pipelines that involve segmentation, layout analysis, and line detection. Another important direction is to explore few-shot and semi-supervised learning to generalize GeezNet across related Ethiopic scripts, extending its impact beyond numerals. These efforts will make the model more practical and reliable in archival digitization workflows.

Beyond its technical contributions, GeezNet plays a meaningful role in preserving cultural heritage. Ge'ez is one of the world's oldest scripts and remains central to Ethiopian religious and historical identity. Yet much of its recorded legacy—found in manuscripts, scrolls, and inscriptions—remains undigitized and at risk of loss. GeezNet provides a pathway to accelerate digitization efforts and make historical content more accessible to scholars, educators, and the public.

This work demonstrates that AI can serve as a bridge between modern technology and historical preservation. GeezNet offers not only a scalable OCR solution but also a blueprint for developing tailored models for other low-resource scripts. It illustrates how deep learning can support inclusive digitization, ensuring that ancient knowledge is preserved, studied, and shared widely. In doing so, GeezNet contributes to a broader vision of ethical and culturally conscious AI design.

ACKNOWLEDGMENT

This project was made possible through the support of the Undergraduate Research Opportunities Program (UROP) at the University of St. Thomas, MN. Their initiative plays a vital role in fostering and facilitating undergraduate research. The authors acknowledge the use of OpenAI [22] to assist in refining the clarity and readability of the manuscript.

REFERENCES

- [1] G. Gragg, "Ge'ez (ethiopic)," in *The Semitic Languages*, R. Hetzron, Ed. London & New York: Routledge, 1997, pp. 242–260.
- [2] G. Negashe and A. Mamuye, "Modified segmentation algorithm for recognition of older ge'ez scripts written on vellum," *arXiv preprint arXiv:2006.00465*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.00465>
- [3] M. Y. Tadesse, "Amharic handwritten character recognition using deep learning," in *Proc. IEEE Africon*, 2019, pp. 1–6.
- [4] G. Tesfaye and A. Mekonnen, "Optical character recognition for tigrinya script," in *Proc. IEEE Africon*, 2021, pp. 1–5.
- [5] P. Wickramasinghe and A. Fernando, "Handwritten sinhala character recognition using deep learning," in *Proc. IEEE 14th Int. Conf. Industrial and Information Systems (ICIIS)*, 2019, pp. 281–286.
- [6] R. Malhotra and M. T. Addis, "Ge'ez digit recognition model based on convolutional neural network," *Applied Artificial Intelligence*, vol. 38, no. 1, 2024.
- [7] A. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," *arXiv preprint arXiv:1704.04861*, 2017. [Online]. Available: <https://arxiv.org/abs/1704.04861>
- [8] M. Tan and Q. V. Le, "Efficientnet: Rethinking model scaling for convolutional neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114.
- [9] M. Nur, M. Abebe, and R. S. Rajendran, "Handwritten ge'ez digit recognition using deep learning," *Applied Computational Intelligence and Soft Computing*, vol. 2022, p. 12, Nov. 2022.
- [10] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [11] G. Cohen, S. Afshar, J. Tapson, and A. V. Schaik, "Emnist: Extending mnist to handwritten letters," *arXiv preprint arXiv:1702.05373*, 2017. [Online]. Available: <https://arxiv.org/abs/1702.05373>
- [12] H. L. Yimer, H. Degefa, M. Cristani, and F. Cunico, "Learning-based ge'ez character handwritten recognition," *arXiv preprint arXiv:2411.13350*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.13350>
- [13] H. A. Aly and M. M. Dessouky, "Ocr for arabic scripts: Challenges and techniques," *Journal of King Saud University - Computer and Information Sciences*, vol. 27, no. 1, pp. 1–15, Jan. 2015.
- [14] S. Sharma, S. S. Sundaram, and C. V. Jawahar, "Online handwritten devanagari character recognition using recurrent neural networks," in *Proc. Int. Conf. Document Analysis and Recognition (ICDAR)*, 2015, pp. 392–396.
- [15] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020. [Online]. Available: <https://arxiv.org/abs/2010.11929>
- [16] C. Li, J. Xu, and Y. Liu, "Transformer-based text recognition for scene images," *Pattern Recognition*, vol. 121, p. 108234, Jan. 2022.
- [17] A. Gupta and S. Singh, "A comprehensive survey of ocr research for complex and underrepresented scripts," *ACM Computing Surveys*, vol. 54, no. 4, pp. 1–35, 2021.
- [18] B. Negash and A. Alemu, "Ge'ez numeral recognition in historical manuscripts using hybrid image processing techniques," *Ethiopian Journal of Science and Technology*, vol. 13, no. 2, pp. 119–132, 2020.
- [19] C. Reul, U. Springmann, C. Wick, and F. Puppe, "Ocr4all: An open-source tool providing a (semi-) automatic ocr workflow for historical printings," in *International Conference on Document Analysis and Recognition (ICDAR)*, 2019, pp. 417–424.
- [20] A. Bharath and S. Madhvanath, "Ocr in indian languages: A survey," in *Proceedings of the International Conference on Frontiers in Handwriting Recognition (ICFHR)*, 2014, pp. 345–350.
- [21] Pakeeza-A, "GeezNet: A lightweight cnn for ge'ez numeral recognition," <https://github.com/Pakeeza-A/GeezNet>, 2025, accessed: Jan. 2025.
- [22] OpenAI, "Gpt-4," <https://chat.openai.com/>, San Francisco, CA, USA, 2023, accessed: 2025-08-13.